

ConDTC: Contrastive Deep Trajectory Clustering for Fine-grained Mobility Pattern Mining

Junjun Si, Jin Yang, Yang Xiang, Li Li, Bo Tu, Rongqing Zhang

Abstract—Trajectory clustering is a cornerstone task in the field of trajectory mining. With the proliferation of deep learning, deep trajectory clustering has been widely researched to mine mobility patterns from massive unlabeled trajectories. Nevertheless, existing methods mostly ignore trajectories' temporal regularities, which are essential for mining fine-grained mobility patterns for applications including traveling group identification, transportation mode discovering, social security emergency, etc. To fill this gap, we propose ConDTC, a contrastive deep trajectory clustering method targeting for fine-grained mobility pattern mining. Specifically, we first design a spatial-temporal trajectory representation learning method which can capture both spatial and temporal regularities of trajectories synchronously. The proposed trajectory representation model can be used as a pre-trained model to serve various downstream trajectory mining tasks. Then, we construct a contrastive trajectory clustering module which optimizes trajectory representations and clustering performance simultaneously. Experimental results on three datasets validate that ConDTC can identify fine-grained mobility patterns by clustering trajectories with similar spatial-temporal mobility patterns together while separating those with different mobility patterns apart. Actually, ConDTC outperforms all state-of-the-art competitors substantially in terms of effectiveness, efficiency and robustness. The source code is available at <https://github.com/TrajResearch/ConDTC>.

Index Terms—trajectory clustering, trajectory representation learning, trajectory mining, spatial-temporal data mining.

1 INTRODUCTION

WITH the proliferation of mobile communication devices and sensors, a massive amount of trajectories, containing rich mobility information of human and objects, have been generated [1], [2], [3]. Since labeling massive trajectories is usually difficult, trajectory clustering has been a fundamental task to mine mobility patterns from vast amount of unlabeled trajectories in real-world applications including traffic monitoring, abnormal behavior detection, activity understanding, etc.

In the literature, a plethora of trajectory clustering methods have been proposed and can be categorized into two kinds: traditional trajectory clustering and deep learning based trajectory clustering [4]. Traditional trajectory clustering methods focus on the problem of quantifying trajectory similarities and then apply well-known clustering algorithms such as K-means to the trajectory data [5], [6], [7]. However, raw trajectories are usually non-uniform and variable, causing traditional trajectory clustering methods computationally complex and susceptible to noises. Therefore, deep-learning based trajectory clustering has emerged.

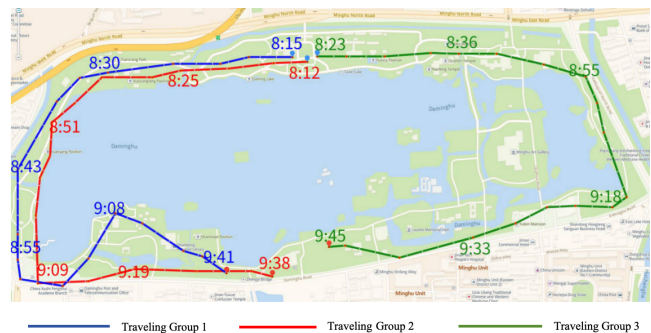


Fig. 1. Traveling group identification scenario: there are three traveling groups, but coarse-grained trajectory clustering can hardly identify group 1 and group 2 because the temporal information of trajectories are neglected and these two groups share similar moving paths. Consequently, fine-grained trajectory clustering considering temporal regularities as well as spatial regularities for fine-grained mobility pattern mining is necessary.

It casts non-uniform trajectories to fixed-length representation vectors via deep-learning models and clusters the representation vectors instead [8], [9], [10], [11]. Nevertheless, current deep-learning based trajectory clustering methods mostly focus on the spatial trajectories, neglecting the temporal features when each location is visited. As a result, fine-grained mobility pattern mining cannot be addressed by existing methods, failing to serve fine-grained trajectory mining applications [12], [13]. We refer the existing trajectory clustering methods that only consider spatial features as coarse-grained trajectory clustering, which can be utilized for mining hot routes, frequent mobility patterns, abnormal activities, etc. And we aim to propose a fine-grained trajectory clustering method that considers both

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spatial and temporal features to identify traveling together groups, discovering transportation modes, etc.

Take the traveling group identification problem as an example, groups of visitors may start their tour along several preset routes in a scenic spot, and groups on the same route may have different temporal regularities, such as group 1 and group 2 in Fig. 1. Therefore, it requires fine-grained trajectory clustering, which takes the temporal information into consideration as well as the spatial information, to mine fine-grained mobility patterns.

To solve the problem of fine-grained trajectory clustering, we need to overcome two main challenges. On the one hand, how to capture the fine-grained mobility patterns? On the other hand, how to cluster trajectories with similar spatial-temporal patterns together while separating those with different spatial-temporal patterns apart?

Trajectory representation learning is usually utilized to solve the first challenge. Early methods generally focused on the spatial sequences of trajectories and utilized auto-encoders to obtain trajectory representation vectors [14], [15], [16], neglecting the temporal sequences of trajectories. To deal with this issue, Fu [17] and Fang [9] employed metric learning to capture the spatial and temporal features of trajectories jointly. However, the performance of these models are limited by the trajectory similarity metrics employed to train the models. Further, Jiang crafted START [18] which reformed the Transformer encoder and utilized contrastive learning to learn spatial and temporal features via the masked language model. However, START mainly learns the temporal features through contrastive learning because the masked language model only predicts the spatial tokens.

As for the second challenge, early studies apply traditional clustering methods such as K-means on trajectory representation vectors [8], [10], [11], [19]. However, the clustering performance of these two-stage methods is limited by splitting the trajectory representation and clustering process as two separate parts. Therefore, researchers then developed methods in an end-to-end manner to optimize the trajectory representations and clustering performance simultaneously [4], [9]. Nevertheless, the clustering performance is mainly determined by the quality of the trajectory representation vectors.

To solve the above challenges, we introduce ConDTC, a contrastive deep trajectory clustering method for fine-grained mobility pattern mining. **Firstly, a novel Spatial-Temporal Trajectory representation model, STTraj2Vec, is designed to capture fine-grained mobility patterns of trajectories while translating them into low-dimensional vectors. With STTraj2Vec, we incorporate the spatial and temporal regularities of trajectories by modifying the BERT model. Secondly, a contrastive trajectory clustering module is devised to optimize the trajectory representation vectors and clustering performance simultaneously. We utilize contrastive learning to enhance the similarity of similar trajectories and enlarge the dissimilarity of dissimilar trajectories.** Thirdly, extensive experiments have been conducted with three trajectory datasets. And ConDTC demonstrates significant performance improvement compared to state-of-the-art methods with respect to effectiveness, efficiency and robustness.

In summary, the main contributions of our work are summarized as follows:

- We propose ConDTC to address the trajectory clustering problem for fine-grained mobility pattern mining. It can learn the fine-grained mobility patterns of trajectories through a novel trajectory representation model and contrastive trajectory clustering module.
- We design a novel trajectory representation model, which captures the spatial and temporal features of trajectories synchronously by incorporating the spatial and temporal information with a crafted masked spatial-temporal model. In addition, it is efficient and adaptable to support a wide range of trajectory mining tasks as a pre-training model.
- We craft a contrastive trajectory clustering module, which optimizes the trajectory representation vectors and the clustering performance simultaneously. It augments trajectories and enhances the model to learn more fine-grained mobility patterns by a well-designed contrastive loss.
- We conduct extensive experiments with three trajectory datasets. Experimental results show that ConDTC can effectively solve the fine-grained trajectory mining problem with massive unlabeled trajectories. In addition, it shows significant robustness and efficiency compared to the existing state-of-the-art approaches.

2 RELATED WORKS

Existing related works on trajectory clustering can be divided into two types: traditional trajectory clustering based on manual trajectory similarities and deep learning based trajectory clustering.

2.1 Traditional Trajectory Clustering

Traditional trajectory clustering methods mainly consist of two modules: trajectory similarity measure and clustering [6], [20]. Despite the widely used euclidean distance [21], [22], [23], researchers developed various similarity measurements based on manually defined features, such as Dynamic Time Warping (DTW) [24], [25], Edit Distance on Real sequences (EDR) [26], Edit Distance with Projection (EDwP) [27], Edge-Based Distance (EBD) [28], etc. Then, classic clustering methods such as Density Based Spatial Clustering of Applications with Noise (DBSCAN), K-means, and hierarchical clustering are applied to separate trajectories [5], [29]. However, manual trajectory similarities often lead to quadratic computational complexity and are generally not applicable for a wide variety of data and applications. Further, some works [29], [30] got snapshot clusters of trajectory points at each time point first, which results in high computations as well.

2.2 Deep Learning based Trajectory Clustering

Deep learning has been widely used in trajectory mining tasks [31], [32]. And deep trajectory clustering mainly consists of two parts: trajectory representation learning and the

clustering process. Early studies focused on the trajectory representation learning and then applied traditional clustering methods to the learned vectors [8], [10], [11], [19]. Recently, researchers have proposed end-to-end trajectory clustering methods which optimizes the trajectory representations for better clustering performance [4], [9].

2.2.1 Trajectory Representation Learning

Trajectory representation learning converts raw trajectories to fixed-length low-dimensional vectors for analysis. Yao [8] used a sliding window to extract moving features of trajectories and then employed a sequence-to-sequence auto-encoder to transform the features to fixed-length vectors. However, the manually defined trajectory features are susceptible to noises. Li [14] reconstructed high-sampling trajectories from noisy low-sampling trajectories to obtain robust representations. Further, Yao [15] employed metric learning to get trajectory representation by approximating some trajectory similarity function. To further improve the quality of trajectory representations, Zhang [33] designed sub-trajectory distance supervision. Yang [16], Zhang [19] and Wu [34] utilized attention schemes for better performance. With the rise of contrastive learning, Liu [35] and Mao [36] trained trajectory representation models via contrastive learning.

Despite the efforts of these studies, they all ignore the temporal sequences of trajectories and are inadequate for fine-grained mobility pattern mining. Therefore, Fu [17] proposed Trembr by incorporating the travel time and road segment information into the trajectory representations jointly. Fang [37] proposed ST2Vec which encoded the temporal sequences of trajectories and utilized trajectory similarity metric learning to obtain trajectory representations. However, Trembr does not capture the periodic patterns and ST2Vec requires extensive computation in the preprocessing process to calculate the metric. Later, Jiang crafted START which [18] reformed the Transformer encoder and utilized contrastive learning to better learn spatial and temporal features of trajectories simultaneously.

2.2.2 End to End Deep Trajectory Clustering

Fang [9] proposed E²DTC to cluster trajectories by jointly optimizing the trajectory reconstruction loss and the clustering loss. Besides, Wang [4] proposed a similar method named DTC. They both employed RNN-based auto encoders to obtain trajectory representations and set the clustering centroids as trainable parameters via self-training.

To the best of our knowledge, there exists no self-training trajectory clustering methods for fine-grained mobility pattern mining applications including traveling group identification, transportation mode discovering and social security emergency. To fill this gap, we propose ConDTC, which solves the problem **by crafting a novel spatial-temporal trajectory representation model and a contrastive clustering module.**

3 PRELIMINARIES

This section presents key definitions related to deep trajectory clustering.

Definition 1 (Trajectory). According to [20], a trajectory T can be defined as a time ordered location-ID sequence, represented as:

$$T = \{p_1, p_2, \dots, p_i, \dots, p_n\} \quad (1)$$

where p_i is the i -th trajectory point, represented as $p_i = (loc_i, ts_i)$, where loc_i is the location-ID, ts_i is the timestamp when the trajectory point is recorded, n is the total number of trajectory points recorded.

Notation (Trajectory Representation Learning). Trajectory representation learning is to transform variable raw trajectories into fixed-length low-dimensional representation vectors while maintaining each trajectory's spatial-temporal features for trajectory mining.

Notation (Deep Trajectory Clustering). Given a set of trajectories S , deep trajectory clustering aims to obtain high-quality trajectory representation vectors and cluster similar trajectories into the same clusters and different trajectories into different clusters.

Notation (Coarse-grained Trajectory Clustering). Most existing trajectory clustering methods neglect the temporal information of trajectories and the mined mobility patterns contain only spatial regularities. We classify those methods as coarse-grained trajectory clustering since they can only cluster trajectories with similar moving path. Coarse-grained trajectory clustering are widely applied to mine hot routes, frequent mobility patterns, abnormal activity prediction, etc.

Notation (Fine-grained Trajectory Clustering). The visiting time sequences of trajectories are informative too. Therefore, we define fine-grained trajectory clustering considering both spatial and temporal features of trajectories. Fine-grained trajectory clustering can serve a large number of applications including traveling group identification, transportation mode discovering, social security emergency and so on.

Problem Statement. The purpose of this paper is to design a deep trajectory clustering model f_c to obtain fine-grained trajectory representation vectors and set the same cluster labels for spatially and temporally similar trajectories, as defined in equation (2):

$$(cluster_i, Z_i) = f_c(T_i) \quad (2)$$

where $cluster_i$, Z_i are the clustering label and d^Z -dimensional trajectory representation vector of trajectory T_i , respectively. And f_c represents the deep trajectory clustering model.

Table 1 lists the mathematical notations used in this paper.

4 THE CONDTC MODEL

4.1 Overview

The proposed trajectory clustering model, ConDTC, is composed of two main parts that work together seamlessly to capture fine-grained mobility patterns of trajectories, as shown in Fig. 2. The first part is a novel spatial-temporal trajectory representation model which aims to capture the fine-grained mobility patterns of trajectories. The second part is a novel contrastive trajectory clustering module to cluster trajectories while extracting more fine-grained mobility patterns.

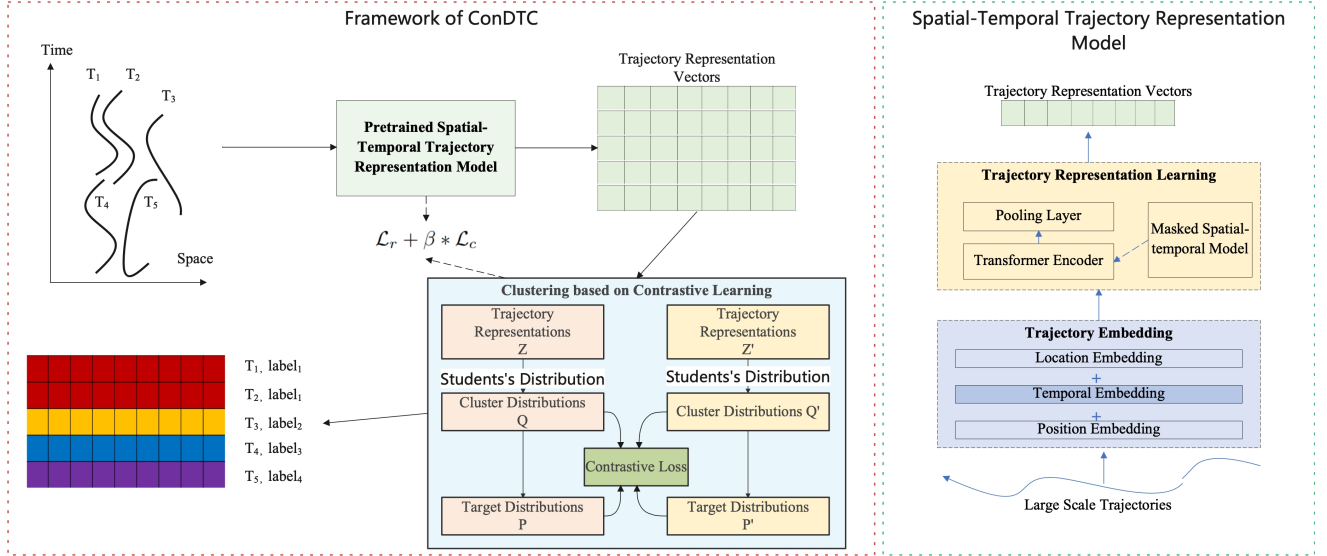


Fig. 2. The framework of ConDTC for fine-grained trajectory clustering. A novel spatial-temporal trajectory representation model and a contrastive trajectory clustering module are proposed to capture spatial and temporal trajectory regularities simultaneously.

TABLE 1
Description of Notation.

Symbol	Description
T	trajectory
p_i	the i -th trajectory point
loc_i	the spatial token of the i -th trajectory point
ts_i	the visiting time of the i -th trajectory point
ts'	the time index of the i -th trajectory point
τ	the trajectory partition period
δ	the time interval
ts'	the time index of the i -th trajectory point
E	embedding vectors of trajectories
d^E	the dimension of embedding vectors
Z	representation vectors of trajectories
d^Z	the dimension of representation vectors
Q	calculated cluster distributions of trajectories
P	target cluster distributions of trajectories
C	the cluster centroids of trajectories
H	representation vectors of trajectory points
θ	the parameter set
α	the hyper-parameter of temporal loss set
β	the hyper-parameter of clustering loss set
σ	the spatial noise parameter
ϵ	the temporal noise parameter

4.2 Spatial-Temporal Trajectory Representation Learning

The architecture of the proposed spatial-temporal trajectory representation model, named STTraj2Vec, is shown in Fig. 2. It consists of two major parts: trajectory embedding and trajectory representation learning. The trajectory representation learning part utilizes the Transformer encoder to capture trajectory mobility patterns bi-directionally and introduces a novel masked spatial-temporal model to train the Transformer encoder.

4.2.1 Trajectory Embedding

The trajectory embedding layer incorporates the spatial and temporal information of the raw trajectories into the Transformer encoder. Firstly, we convert the original trajectory T , defined in equation (1), to a pair of spatial sequence $T_s = \{loc_1, loc_2, \dots, loc_i, \dots, loc_n\}$ and temporal sequence $T_t = \{ts_1, ts_2, \dots, ts_i, \dots, ts_n\}$. Secondly, The timestamps in the temporal sequence is further converted to time indexes:

$$ts'_i = ts_i \% \tau / \Delta \quad (3)$$

where τ represents the trajectory partition period and is set to one day, namely, 86400 seconds in experiments, Δ is the predefined time interval which is set to 60 seconds in experiments. The temporal interval may vary, taking into account factors such as sequence length or movement velocity. Thirdly, **all the spatial and temporal sequences are padded with a special token 'PAD' to the same length L , which is the maximum length of all trajectories.** Then, the spatial and temporal sequences are embedded with trainable linear networks to d^E dimensional vectors E_s and E_t :

$$E_s = W_s T_s + b_s \quad (4)$$

$$E_t = W_t T'_t + b_t \quad (5)$$

where $W_s \in \mathbb{R}^{V_s \times d^E}$, $W_t \in \mathbb{R}^{V_t \times d^E}$, b_s and b_t are trainable parameters, V_s and V_t are the vocabulary size of spatial and temporal tokens. Lastly, we incorporate the sequential relationships of spatial and temporal tokens by utilizing the position encoding method proposed in standard Transformer [38] to d^E dimensional vectors E_p as well. To this end, the embedding vector of a trajectory is represented as:

$$E = E_s + E_t + E_p \quad (6)$$

4.2.2 Trajectory Representation Learning

Motivated by the great success of Transformer in processing sequential data, we utilize the Transformer encoder to capture the trajectory mobility patterns bi-directionally. The

output of the Transformer encoder is the representation vectors H of all trajectory points:

$$H = \text{TransEnc}(E) \quad (7)$$

To obtain the representation vector of the trajectory, we simply design a pooling layer over the Transformer encoder by averaging the representation vectors of all trajectory points:

$$Z = \text{Pooling}(H) \quad (8)$$

Intuitively, we can employ the Masked Language Model (MLM) from BERT [39] to train the trajectory representation model. Nevertheless, trajectory representation is a fundamental task which supports a wide range of coarse-grained and fine-grained trajectory mining tasks. Therefore, a trajectory representation model, which adapts to various trajectory mining applications, is required. In recognition of this, we design the Masked Spatial-Temporal Model (MSTM) to enable the Transformer encoder to capture the mobility patterns with regard to the needs of downstream tasks.

Specifically, we first mask 15% of the input trajectory points, with spatial and temporal tokens be substituted with special tokens 'MASKS' and 'MASKT'. Then, we design two separate feed forward networks to predict the true spatial and temporal tokens:

$$\text{pred}_s = \text{FFN}_1(Z) \quad (9)$$

$$\text{pred}_t = \text{FFN}_2(Z) \quad (10)$$

FFN_1 and FFN_2 have the same architecture with a linear layer followed by a *relu* activation layer, another linear layer and a *softmax* layer to get the predicted tokens.

Finally, the objective of the representation model is designed as the combination of the spatial and temporal predictions:

$$O_{\text{MSTM}} = \arg \max_{\theta} \sum_i \sum_j p(\text{locID}_i, ts'_j | \text{FFN}_1(f_r(E)), \text{FFN}_2(f_r(E))) \quad (11)$$

where θ is the set of trainable parameters, p represents the probability generated by the *softmax* layer, and f_r stands for the proposed trajectory representation model.

To achieve the optimization objective, we utilize the cross entropy loss to measure the predicted errors of the spatial and temporal tokens:

$$\mathcal{L}_r^s = - \sum_{i=1}^M \sum_{j=1}^{V_s} \delta_{i,j} \log(p(\text{loc}_{i,j} | T, \theta)) \quad (12)$$

$$\mathcal{L}_r^t = - \sum_{i=1}^M \sum_{j=1}^{V_t} \delta'_{i,j} \log(p(ts'_{i,j} | T, \theta)) \quad (13)$$

$$\mathcal{L}_r = \mathcal{L}_r^s + \alpha * \mathcal{L}_r^t \quad (14)$$

where α is the hyper-parameter to adjust the model's ability to capture the temporal information of trajectories. δ and δ' are the one-hot representations of ground-truth spatial and temporal tokens. M represents the number of masked tokens.

4.3 Contrastive Trajectory Clustering

The contrastive trajectory clustering module consists of two phases, data augmentation and clustering based on contrastive learning.

4.3.1 Data Augmentation

Contrastive learning aims to maximizing the similarities between similar positive samples while minimizing the similarities between views of dissimilar negative samples.

Therefore, the crucial step in contrastive learning is how to construct positive and negative samples. The generation of trajectory points varies from person to person and from time to time. For example, some people of a traveling group moving together generate records at a place while others do not, and these records are usually not generated at the same time. In recognizing of this, we set the following data augmentation methods:

- *Dropout*: We randomly drop 10% of the trajectory points to create positive samples with different samplings.
- *Time Offset*: We randomly add an offset, which varies in (0, 120) seconds, to the visiting time of each trajectory point. The aim is to mimic the temporal differences between trajectories.

4.3.2 Clustering based on Contrastive Learning

To achieve deep clustering, we design the cluster centers C as trainable parameters with dimension d^Z . Nevertheless, we initialize the cluster center vectors C by applying K-means to the pre-trained trajectory representation vectors Z . The number of clusters K can be determined by the elbow method with K-means. Inspired by t-SNE [40], we leverage the Student's t-distribution with one degree of freedom (which is the same as a Cauchy distribution) to measure the similarity between a trajectory representation vector and the cluster centers. The probability q_{ij} of the trajectory representation vector Z_i belongs to cluster C_j is calculated as:

$$q_{ij} = \frac{(1 + \|Z_i - C_j\|^2)^{-1}}{\sum_k (1 + \|Z_i - C_k\|^2)^{-1}} \quad (15)$$

The Student's t-distribution with one degree of freedom has the particularly nice property that $(1 + \|Z_i - C_j\|^2)^{-1}$ approaches an inverse square law for large pairwise distances $\|Z_i - C_j\|^2$ in the low-dimensional map. This makes the map's representation of joint probabilities almost invariant to changes in the scale of the map. It also means that large clusters of points that are far apart interact in just the same way as individual points, so the optimization operates in the same way at all but the finest scales. Finally, the trajectories are assigned to clusters with the biggest probabilities according to $Q = \{q_{ij}\}$.

Subsequently, we utilize the spatial-temporal trajectory representation model to transform the augmented trajectories into vectors, denoted as Z and Z' . Then, we can get two cluster probability distributions Q and Q' . For Q_i , the trajectory vector Q'_i with the same index in Q' , is viewed as its positive sample, while the left $2 * \text{batchsize} - 2$ vectors are viewed as its negative samples. To achieve self-supervised

clustering, we calculate target distributions P and P' based on the cluster probability distributions Q and Q' [9]:

$$p_{ij} = \frac{q_{ij}^2 / f_j}{\sum_k q_{ik}^2 / f_k}, \quad p'_{ij} = \frac{q'_{ij}{}^2 / f'_j}{\sum_k q'_{ik}{}^2 / f'_k} \quad (16)$$

$$f_k = \sum_i q_{i,k} \quad f'_k = \sum_i q'_{i,k} \quad (17)$$

In the target distributions, each cluster assignment is squared and normalized to pull the trajectory vector closer to the cluster centers, so as to improve the cluster purity.

In [41], the authors defined the contrastive loss as the sum of all positive pairs' loss to learn sentence representations. Motivated by this, we define a contrastive clustering loss as the sum of clustering probability distributions of positive pairs. Specifically, we **achieve the clustering optimization objective by minimizing the Kullback-Leibler divergence loss (KL loss) between the clustering probability distributions and the target probability distributions of different views of trajectories**. To this end, we design the contrastive clustering loss $\mathcal{L}_c$ as:

$$\begin{aligned} \mathcal{L}_c &= (KL(P' \| Q) + KL(P \| Q')) / 2 \\ &= (\sum_i \sum_j p'_{ij} \log \frac{p'_{ij}}{q_{ij}} + \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q'_{ij}}) / 2 \end{aligned} \quad (18)$$

Finally, the cluster layer fine-tunes the spatial-temporal trajectory representation model together with the trajectory clustering performance. **Therefore, the total loss of ConDTC is designed as:**

$$\mathcal{L} = \mathcal{L}_r + \beta * \mathcal{L}_c = \mathcal{L}_r^s + \alpha * \mathcal{L}_r^t + \beta * \mathcal{L}_c \quad (19)$$

where β is the hyper-parameter to adjust the clustering loss ratio.

5 GROUND-TRUTH GENERATION

To the best of our knowledge, there are no publicly available trajectory datasets with labels for fine-grained mobility pattern mining. To this end, we design a ground truth generation algorithm, which allows researchers to generate trajectory clusters with a few base trajectories, as shown in Algorithm 1. *addLocNoise* is implemented by adding a random Gaussian noise $\sim \mathcal{N}(0, \sigma)$ to each location according to [14]. We amplified the noise deviation σ from 30 meters in [14] to 50 meters to ensure the model is more robust to noisy data. *addTimeNoise* is implemented by adding a random noise between $[-\epsilon, \epsilon]$ minutes and make sure the visiting time is monotonically increasing. The time noise value range is related to the time interval and trajectory partition period that are considered as co-movements. ϵ is set to 3 minutes in the following experiments. The base trajectories of Qingdao are shown in Fig. 3.

With each base trajectory, it generates three corresponding trajectory clusters which share the same moving path, but different visiting time or moving speeds by Algorithm 1. The generated clusters have the following attributes for fine-grained trajectory clustering:

- Each cluster represents a group of moving together trajectories. We randomly drop trajectory points, add

spatial and temporal noises to make the trajectories close to realistic accompanying trajectories.

- The clusters share similar moving path but different visiting time. The aim is to simulate groups with different preferences. For example, a senior group and a junior group walking along the lakeside may stop at different spots.
- There are clusters which share similar moving path but different moving speeds. The aim is to simulate groups with different transportation modes. For example, visitors on foot and visitors taking tour cars.

Algorithm 1: Groud Truth Generation Algorithm for Fine-grained Trajectory Clustering

input : A set of base trajectories

$DT = \{T_1, T_2, \dots, T_i, \dots, T_m\}$,
maximum and minimum cluster size
 $maxN, minN$, variation rate p

output: A set of trajectory groups

$DT' = \{G_1, G_2, \dots, G_i, \dots, G_{3m}\}$

```

1   $DT' = \{\}$ ;
2  for  $i \leftarrow 0$  to  $m$  do
3      for  $j \leftarrow 0$  to 3 do
4           $N = \text{randint}(minN, maxN)$ ;
5           $G = \{\}$ ;
6          for  $k \leftarrow 0$  to  $N$  do
7               $T'_i = \text{randomDrop}(T_i, p)$ ;
8               $T'_i = \text{addLocNoise}(T'_i, p)$  #to add a
                  Gaussian noise  $\sim \mathcal{N}(0, \sigma)$  to each
                  location;
9               $T'_i = \text{addTimeNoise}(T'_i, p)$  #to add a
                  random noise between  $[-\epsilon, \epsilon]$  minutes and
                  make sure the visiting time is
                  monotonically increasing;
10             if  $j==0$  then
11                  $G.add(T'_i)$ 
12             if  $j==1$  then
13                  $T''_i = \text{timeShift}(T'_i, \delta_t)$  #to add or
                  subtract  $\delta_t$  to the temporal sequences;
14                  $G.add(T''_i)$ 
15             if  $j==2$  then
16                  $T''_i = \text{speedScale}(T'_i)$  #to change the
                  temporal sequences by scaling the
                  moving speed;
17                  $G.add(T''_i)$ 
18              $DT'.add(G)$ 

```

6 EXPERIMENTS

We conduct experiments to verify the effectiveness of the proposed ConDTC method and answer the following questions:

- How is the performance of ConDTC for fine-grained mobility mining compared to various baselines?
- How is the efficiency of ConDTC compared to state-of-the-art methods?

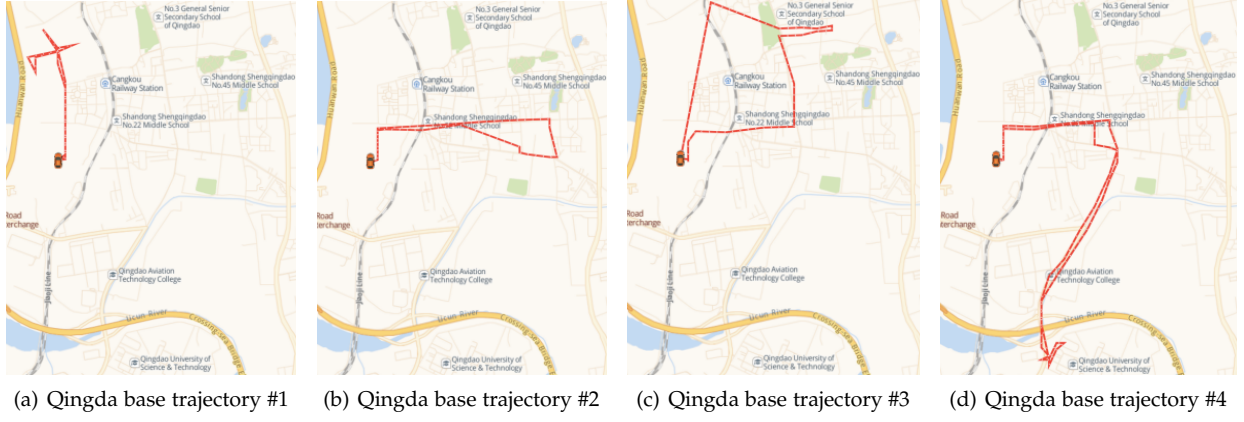


Fig. 3. The base trajectories selected for Qingdao dataset. All base trajectories originate from and end at the Qingdao North Railway Station, and some base trajectories share similar sub-parts to make it more difficult for the clustering methods to discern.

- How is the robustness of ConDTC with respect to pre-training data size and the number of clusters?
- How does each sub-module contribute to the performance of ConDTC?

6.1 Datasets

We utilize three trajectory datasets to validate the trajectory clustering performance of the proposed ConDTC model for fine-grained mobility pattern mining. The descriptions of the datasets are as follows:

- **Porto:**¹ It is an open source taxi trajectory dataset. We randomly selected 9 base trajectories from the original data and utilized Algorithm 1 to generate ground-truth clusters.
- **Qingdao:** It is a taxi trajectory dataset logged by monitors on roads in Qingdao, China. We randomly selected 4 base trajectories and obtained 12 clusters with Algorithm 1.
- **Mobile:** We collected 4 base trajectories through a mobile application installed in our mobile phones, and extended them to clusters with Algorithm 1.

The coordinates in all trajectories are mapped to location IDs with the popular geocoding system Geohash with the length set to 7 for Porto and 8 for the others. Finally, the summary of datasets is shown in Table 2.

6.2 Baselines

To our best knowledge, no previous trajectory clustering research address the fine-grained mobility pattern mining problem. Therefore, we set the following models as baselines to evaluate the superiority of ConDTC.

- **E²DTC:** It is an end-to-end deep trajectory clustering framework via self-training without manual feature extractions, focusing on the spatial sequences of trajectories [9].
- **SCCL:** It is a text clustering method based on contrastive learning. Since the location sequences of trajectories can be viewed as texts, we train SCCL with

trajectories to verify the effectiveness of contrastive learning on clustering [42].

- **CLTSim+K-means:** CL-TSim [43] is trajectory representation learning model based on contrastive learning. We utilize CL-TSim to obtain trajectory representation vectors and then apply K-means for clustering.
- **BERT+K-means:** BERT is effective to model sequential data. Therefore, we utilize it to obtain trajectory representation vectors and then apply K-means for clustering.
- **START+K-means:** START [18] is the state-of-the-art unsupervised trajectory representation method which combines the spatial and temporal features of trajectories. We utilize START to obtain fine-grained trajectory representation vectors and then apply K-means for clustering.
- **ST2Vec+K-means:** ST2Vec [37] is the state-of-the-art supervised trajectory representation method combining the spatial and temporal features of trajectories via metric learning. We utilize ST2Vec to obtain fine-grained trajectory representation vectors and then apply K-means for clustering. Specifically, we select the TP [44] metric for supervision since it shows the best performance in ST2Vec’s reported results.

6.3 Experiment Settings

ConDTC is implemented by Python 3.7 and PyTorch (1.11.0+cu113). We utilize the AdamW optimizer to train the ConDTC model and run experiments on a server with two Tesla T4 GPU (16G Memory). The sizes of trajectory embedding vectors d^E and d^Z are both set to 256 for a fair comparison with START. The batch size is set to 32. The pre-training and fine-tuning learning rates of STTraj2Vec are set to $15e-5$, $5e-5$, and the clustering layer learning rate is set to $15e-5$ as well.

6.4 Evaluation Metrics

To evaluate the clustering performance, we employ three widely used metrics: Unsupervised clustering Accuracy (UACC), Normalized Mutual Information (NMI), and Rand Index (RI) as introduced in [9]. It should be noted that NMI and RI are biased in some circumstances. For example, if

1. <https://www.kaggle.com/c/pkdd-15-predict-taxi-service-trajectory-i/data>

TABLE 2
Dataset Summary.

Dataset	Trajectory Numbers	Cluster Numbers	Distinct Trajectory Points	Distinct Location IDs
Porto	27009	27	397908	368
Qingdao	12004	12	91480	144
Mobile	11604	12	192489	342

the clustering model cannot distinguish trajectories between two clusters, it will generate high NMI and AI values while the UACC value is low.

$$UACC = \max_m \frac{\sum_{i=1}^N \mathbf{1}\{C'_i = m(C_i)\}}{N} \quad (20)$$

where N is the total number of trajectories, C_i and C'_i are the generated and ground-truth cluster of trajectory T_i , $m()$ represents Hungarian Algorithm which maps the generated cluster label with the ground-truth.

$$NMI(C, C') = \frac{I(C, C')}{\sqrt{H(C)H(C')}} \quad (21)$$

where $H()$ calculates the entropy, $I()$ denotes the mutual information.

$$RI = \frac{TP + TN}{N(N - 1)/2} \quad (22)$$

where $TP(TN)$ denotes the number of trajectory pairs where trajectories from the same(different) classes in the ground-truth dataset are also classified into the same(different) classes in prediction.

6.5 Experimental Results

6.5.1 Performance Comparison

Table 3 shows the performance of the ConDTC and baseline models. We give our observations and corresponding analysis as follows:

(1) ConDTC achieves accurate clustering on all datasets. The superiority of ConDTC attributes to the proposed spatial-temporal trajectory representation model and the contrastive trajectory clustering module. On the one hand, the spatial-temporal trajectory representation model captures fine-grained mobility patterns by learning the spatial and temporal sequences of trajectories simultaneously. On the other hand, the contrastive trajectory clustering layer further fine-tunes the trajectory representations for better clustering performance utilizing contrastive learning.

(2) As a supervised model, ST2Vec demonstrates the best clustering results among all baselines. ConDTC outperforms it by 11% improvement on Porto and shows comparable results with it on Qingdao. However, it fails to cluster the Mobile dataset with all four trajectory similarity measures embedded. It verifies that the generalization of ST2Vec is limited by the selected metric to train the model.

(3) START is able to capture both spatial and temporal mobility patterns as well. However, START doesn't set training goals for temporal information directly, affecting its ability to distinguish fine-grained trajectory patterns. From experiments, we find that START mainly captures the temporal information via contrastive learning.

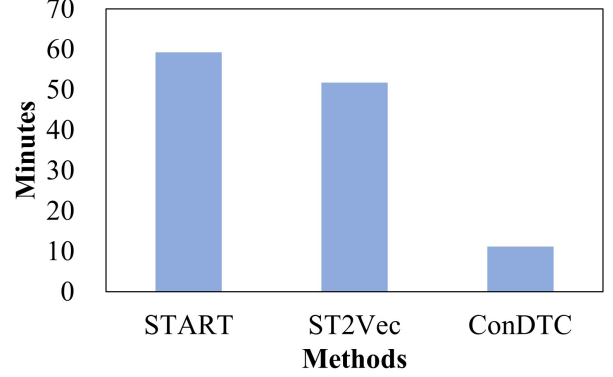


Fig. 4. Efficiency comparison of ConDTC, START+K-means and ST2Vec+K-means on Porto.

(4) Methods including E²DTC, SCCL, CL-TSim+Kmeans and BERT+K-means ignore the temporal sequences of trajectories, and thus fail to capture fine-grained mobility patterns. The clustering accuracy of these methods is around 0.33. Because for each cluster, there are two other clusters that share similar spatial paths with it.

6.5.2 Efficiency Study

From Fig. 4, we can tell ConDTC has significant efficiency superiority compared to the best two baselines. ConDTC reaches excellent clustering performance within 12 minutes. ST2Vec is a trajectory representation method based on metric learning while ConDTC is unsupervised. ST2Vec is computational complex because it requires to compute pairwise trajectory similarities with a selected trajectory similarity measurement to train the model. The proposed ConDTC and START are both based on BERT. The reason why START is time consuming is that START employs contrastive learning for trajectory representations. When we turnoff the contrastive loss of START, it becomes as efficient as the proposed ConDTC.

6.5.3 Sensitivity Study

We perform sensitivity study to evaluate the robustness of ConDTC. Specifically, we evaluate the effects of pre-training data size, number of clusters as well as hyper-parameters in equation (19).

Sensitivity to pre-training data size. We randomly select subsets of the datasets for pre-training the spatial-temporal trajectory representation model with the pre-training data size ratio varies in (50%, 60%, 70%, 80%, 90%). The UACC values on Porto is demonstrated in Fig. 5(a). ConDTC demonstrates high robustness with varying pre-training data size, while the results of START and ST2Vec fluctuates largely. START employs contrastive learning to learn the

TABLE 3
Clustering Performance of ConDTC and Baselines.

Model	Dataset	Porto			Qingdao			Mobile		
		UACC	NMI	RI	UACC	NMI	RI	UACC	NMI	RI
E ² DTC		0.3340	0.6785	0.3413	0.3388	0.5894	0.3388	0.3376	0.6719	0.4114
SCCL		0.3389	0.6878	0.3603	0.3300	0.5667	0.2918	0.3326	0.5763	0.3103
CL-TSim+K-means		0.3459	0.7464	0.4327	0.3397	0.6437	0.3929	0.3346	0.6274	0.3854
BERT+K-means		0.3411	0.6766	0.3354	0.3455	0.5653	0.2910	0.3424	0.5690	0.2962
ST2Vec+K-means		0.8987	0.9497	0.8740	0.9835	0.9781	0.9663	—	—	—
START+K-means		0.8217	0.9279	0.8151	0.8332	0.9042	0.8055	0.8817	0.9461	0.8765
ConDTC		0.9976	0.9973	0.9951	0.9829	0.9775	0.9650	0.9912	0.9885	0.9818

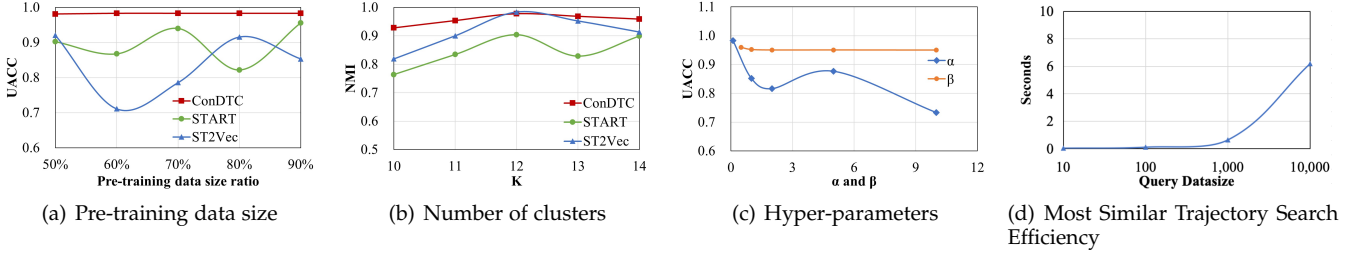


Fig. 5. Sensitivity study: (a) shows the effect of pre-training data size; (b) shows the sensitivity to the number of clusters; (c) shows the hyper-parameter tuning results; (d) shows the most similar trajectory search efficiency from 100,000 trajectories based learned trajectory vectors.

trajectory representations, in which the quality of negative samples varies with the input data. Similarly, ST2Vec creates similar and dissimilar pairs of trajectories for training, the quality of trajectory pairs determines the performance of the model.

Sensitivity to the number of clusters. The number of clusters K are directly given in the above experiments to evaluate the clustering performance. In real-world applications, the optimal number of clusters can be determined by the elbow method [45]. Nevertheless, will the clustering performance be significantly affected if the selection of K is incorrect? To answer this question, we evaluate the sensitivity of ConDTC to the number of clusters on dataset Qingdao. As the ground-truth K of Qingdao is 12, we vary the input K of ConDTC in (10, 11, 12, 13, 14) and demonstrate the results in Fig. 5(b). As observed, the fluctuation of the NMI values of ConDTC is much smaller compared to START and ST2Vec. ConDTC remain stable even if the input K is different with the ground-truth value.

Sensitivity to hyper-Parameters We conduct experiments to determine the high-level parameters in the total loss function, namely, α and β in equation (19). The $UACC$ values with different settings are demonstrated in Fig. 5(c). The figure shows that α significantly affects the clustering performance of ConDTC, and the $UACC$ tops with α set to 0.1. Obviously, the value of α shall be determined considering the objectives of tasks. If the aim is to mine hot route from trajectories, α should be set to zero. On the contrary, the value of β has much smaller effects on the clustering performance. According to the test results, we set β to 0.5.

6.5.4 Ablation Study

To verify the effectiveness of each proposed component of ConDTC, we conduct ablation experiments with the following variations:

TABLE 4
Ablation study results of ConDTC.

Model	UACC	NMI	RI
ConDTC	0.9976	0.9973	0.9951
w/o $\mathcal{L}_{kl}$	0.9101	0.9374	0.8443
w/o DataAug	0.9803	0.9811	0.9614
w/o E_t	0.3484	0.6789	0.3390
w/o $\mathcal{L}_r^t$	0.3554	0.6742	0.3280

- **w/o $\mathcal{L}_{kl}$:** We turned off the cluster layer of ConDTC and applied K-means for clustering.
- **w/o DataAug:** We turned off the data augmentation schemes in subsection 4.3.1.
- **w/o E_t :** We turned off the temporal embedding, in which case only the spatial sequences of trajectories are considered.
- **w/o $\mathcal{L}_r^t$:** We turned off the temporal loss function in equation (14) to verify our assumption that temporal embedding is not sufficient to learn the temporal features of trajectories.

We have the following observations from the ablation experimental results on Porto shown in Table 4. First, the comparable results of w/o E_t and w/o $\mathcal{L}_r^t$ indicates that the temporal embedding scheme is insufficient to extract temporal features of trajectories. Second, the data augmentation schemes are effective to improve the clustering performance. Because the data augmentation considering realistic accompanying trajectory features provides better views of trajectories. Third, the result of w/o $\mathcal{L}_{kl}$ implies that the contrastive trajectory clustering layer is crucial to significantly improve the performance of ConDTC. Because with contrastive learning, ConDTC can further learn more fine-grained mobility patterns of trajectories.

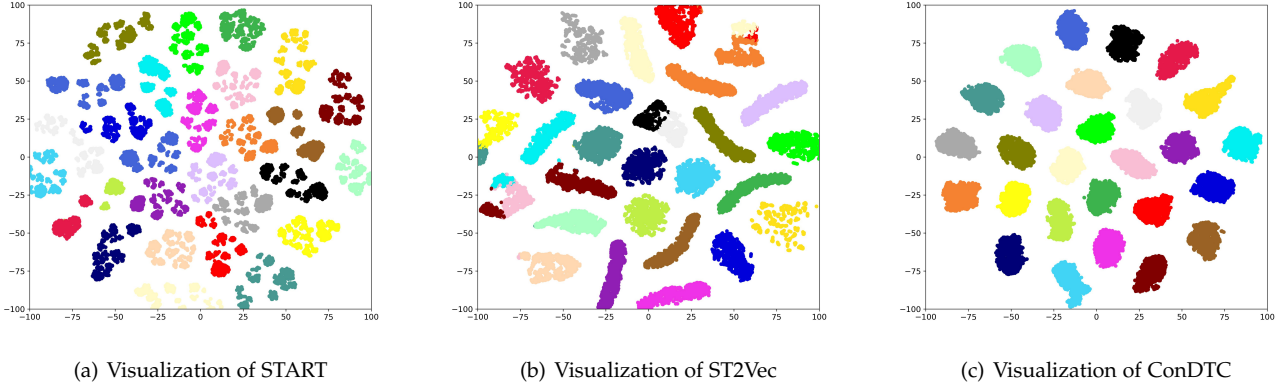


Fig. 6. Visualization on Porto: (a), (b) and (c) show t-SNE visualization of learned trajectory representations by START, ST2Vec and ConDTC, respectively.

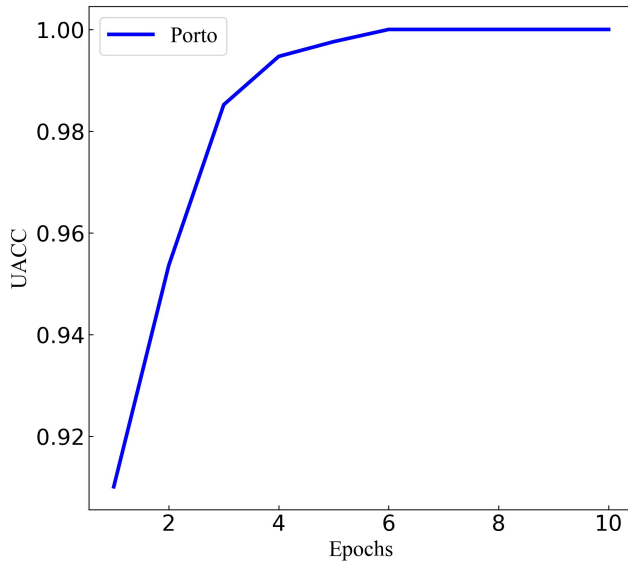


Fig. 7. The UACC curve of ConDTC versus clustering epochs.

6.5.5 Visualization

We utilized the t-SNE method to provide visualizations of clustering results on Porto. As shown in Fig. 6(a) to Fig. 6(c), the clustering purity and the separation of clusters of ConDTC is significantly improved compared to START and ST2Vec. In addition, we visualized the learning process of ConDTC in Fig. 7. It indicates that the clustering layer can reach stable performance within 6 epochs. This further validates the efficiency of ConDTC.

6.5.6 Case Study of The Trajectory Representation Model

Our industrial partner provided us a CDR (Call Detail Record) dataset containing 143,182 trajectories of citizens. We conduct a most similar trajectory search experiment to verify the effectiveness of the proposed spatial-temporal trajectory representation model. Firstly, we used 80% of the data to train the proposed STTraj2Vec model and two baselines. Secondly, we randomly selected 10,000 trajectories from the remaining 20% data as a query dataset, and created the corresponding ground-truth most similar trajectory

TABLE 5
Most similar trajectory search results of trajectory representation models.

Model	Metrics		
	MR	MRR	HR@1
ST2Vec	38.7090	0.3528	0.2281
START	1.0334	0.9833	0.9666
BERT	2.9331	0.5653	0.2670
proposed	1.0093	0.9954	0.9907

dataset by randomly drop 10% trajectory points from the query. Thirdly, we fused the ground-truth dataset with another 100,000 trajectories as a trajectory database. Finally, we converted the trajectories to vectors and conducted most similar trajectory search. The results are presented in Table 5, *MR*, *MRR* and *HR@1* represents mean rank, mean rank ratio and hit ratio of the ground-truth.

The results in Table 5 implies that the proposed trajectory representation model, STTraj2Vec, represents the spatial-temporal features of trajectories with the highest quality. Its most similar trajectory search hit ratio (*HR@1*) is 0.99, which has 2.5% improvement compared to START. On the contrary, the standard BERT can hardly recognize the ground-truth. Surprisingly, ST2Vec shows the worst performance on this task. We infer it's because the CDR data have significantly more noisy data than GPS trajectories in Porto or Qingdao. CDR records are generated while people uses the mobile services. And the locations in CDR come from Cell-ID positioning, which has large positioning errors. In addition, CDR data are usually much sparser than GPS trajectories. These properties will cause distortions in ST2Vec's trajectory similarity calculation, which is based on point-wise distances. From Fig.5(d), we can tell that trajectory representation makes downstream tasks such as most similar trajectory search much more efficient.

7 CONCLUSION

We introduce ConDTC, a novel trajectory clustering model for mining fine-grained mobility patterns from large-scale trajectory data. Part of the model can be pre-trained on large-scale trajectories to translate original trajectories to

vectors, supporting various trajectory mining tasks including trajectory prediction, travel time estimation, similar trajectory search, etc. Therefore, we believe the work has high research relevance for the development and research of future urban big data. In the future, we plan to explore how to cluster trajectories from different sources, such as APP check-ins, vehicle GPS devices, call detail records, etc.

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REFERENCES

- [1] S. Wang, Z. Bao, J. S. Culpepper, and G. Cong, "A Survey on Trajectory Data Management, Analytics, and Learning," *ACM Computing Surveys*, vol. 54, no. 2, pp. 1–36, Mar. 2022.
- [2] Q. Gao, F. Zhou, K. Zhang, G. Trajcevski, X. Luo, and F. Zhang, "Identifying Human Mobility via Trajectory Embeddings," in *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence*. Melbourne, Australia: International Joint Conferences on Artificial Intelligence Organization, Aug. 2017, pp. 1689–1695.
- [3] A. D. Alowayr, L. A. Alsaloohi, A. M. Alshahrani, and J. Akaichi, "A Review of Trajectory Data Mining Applications," in *2021 International Conference of Women in Data Science at Taif University (WiDSTaif)*, Mar. 2021, pp. 1–6.
- [4] C. Wang, F. Lyu, S. Wu, Y. Wang, L. Xu, F. Zhang, S. Wang, Y. Wang, and Z. Du, "A deep trajectory clustering method based on sequence-to-sequence autoencoder model," *Transactions in GIS*, vol. 26, no. 4, pp. 1801–1820, 2022.
- [5] J. Tang, L. Liu, J. Wu, J. Zhou, and Y. Xiang, "Trajectory clustering method based on spatial-temporal properties for mobile social networks," *Journal of Intelligent Information Systems*, vol. 56, no. 1, pp. 73–95, Feb. 2021.
- [6] J. Bian, D. Tian, Y. Tang, and D. Tao, "A survey on trajectory clustering analysis," *arXiv:1802.06971 [cs]*, Feb. 2018, arXiv: 1802.06971.
- [7] C.-C. Hung, W.-C. Peng, and W.-C. Lee, "Clustering and aggregating clues of trajectories for mining trajectory patterns and routes," vol. 24, no. 2, pp. 169–192. [Online]. Available: <http://link.springer.com/10.1007/s00778-011-0262-6>
- [8] D. Yao, C. Zhang, Z. Zhu, J. Huang, and J. Bi, "Trajectory clustering via deep representation learning," in *2017 International Joint Conference on Neural Networks (IJCNN)*, 2017, pp. 3880–3887, iSSN: 2161-4407.
- [9] Z. Fang, Y. Du, L. Chen, Y. Hu, Y. Gao, and G. Chen, "E2DTC: An End to End Deep Trajectory Clustering Framework via Self-Training," in *2021 IEEE 37th International Conference on Data Engineering (ICDE)*, Apr. 2021, pp. 696–707, iSSN: 2375-026X.
- [10] M. Yue, Y. Li, H. Yang, R. Ahuja, Y.-Y. Chiang, and C. Shahabi, "DETECT: Deep Trajectory Clustering for Mobility-Behavior Analysis," in *2019 IEEE International Conference on Big Data (Big Data)*, 2019, pp. 988–997.
- [11] H. Cao, F. Xu, J. Sankaranarayanan, Y. Li, and H. Samet, "Habit2vec: Trajectory Semantic Embedding for Living Pattern Recognition in Population," *IEEE Transactions on Mobile Computing*, vol. 19, no. 5, pp. 1096–1108, 2020, conference Name: IEEE Transactions on Mobile Computing.
- [12] X. Huang, M. Li, J. Zhang, L. Zhang, H. Zhang, and S. Yan, "Tourists' spatial-temporal behavior patterns in theme parks: A case study of Ocean Park Hong Kong," *Journal of Destination Marketing & Management*, vol. 15, p. 100411, Mar. 2020. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2212571X19303865>
- [13] Z. Yang, R. Tang, Y. Chen, and B. Wang, "Spatial-Temporal Clustering and Optimization of Aircraft Descent and Approach Trajectories," *International Journal of Aeronautical and Space Sciences*, vol. 22, no. 6, pp. 1512–1523, Dec. 2021. [Online]. Available: <https://doi.org/10.1007/s42405-021-00401-y>
- [14] X. Li, K. Zhao, G. Cong, C. S. Jensen, and W. Wei, "Deep Representation Learning for Trajectory Similarity Computation," in *2018 IEEE 34th International Conference on Data Engineering (ICDE)*. Paris: IEEE, Apr. 2018, pp. 617–628. [Online]. Available: <https://ieeexplore.ieee.org/document/8509283/>
- [15] D. Yao, G. Cong, C. Zhang, and J. Bi, "Computing Trajectory Similarity in Linear Time: A Generic Seed-Guided Neural Metric Learning Approach," in *2019 IEEE 35th International Conference on Data Engineering (ICDE)*, Apr. 2019, pp. 1358–1369, iSSN: 2375-026X.
- [16] P. Yang, H. Wang, Y. Zhang, L. Qin, W. Zhang, and X. Lin, "T3S: Effective Representation Learning for Trajectory Similarity Computation," in *2021 IEEE 37th International Conference on Data Engineering (ICDE)*, Apr. 2021, pp. 2183–2188, iSSN: 2375-026X.
- [17] T.-Y. Fu and W.-C. Lee, "Trembr: Exploring Road Networks for Trajectory Representation Learning," *ACM Transactions on Intelligent Systems and Technology*, vol. 11, no. 1, pp. 1–25, Feb. 2020. [Online]. Available: <https://dl.acm.org/doi/10.1145/3361741>
- [18] J. Jiang, D. Pan, H. Ren, X. Jiang, C. Li, and J. Wang, "Self-supervised Trajectory Representation Learning with Temporal Regularities and Travel Semantics," in *2023 IEEE 39th international conference on data engineering (ICDE)*. IEEE, 2023.
- [19] R. Zhang, P. Xie, H. Jiang, Z. Xiao, C. Wang, and L. Liu, "Clustering Noisy Trajectories via Robust Deep Attention Auto-Encoders," in *2019 20th IEEE International Conference on Mobile Data Management (MDM)*, Jun. 2019, pp. 63–71, iSSN: 2375-0324.
- [20] Y. Zheng, "Trajectory Data Mining: An Overview," *ACM Transactions on Intelligent Systems and Technology (TIST)*, vol. 6, no. 3, pp. 1–41, 2015.
- [21] Y. Yang, J. Cai, H. Yang, J. Zhang, and X. Zhao, "TAD: A trajectory clustering algorithm based on spatial-temporal density analysis," *Expert Systems with Applications*, vol. 139, p. 112846, Jan. 2020. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S0957417419305482>
- [22] W. Wang, Z. Luan, B. He, X. Li, D. Zhang, Z. Huang, and W. Tu, "A NEW HIERARCHICAL CLUSTERING APPROACH FOR SPARSE MOBILE PHONE TRAJECTORIES," *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. XLII-4, pp. 697–701, Sep. 2018.
- [23] M. Nanni and D. Pedreschi, "Time-focused clustering of trajectories of moving objects," *Journal of Intelligent Information Systems*, vol. 27, no. 3, pp. 267–289, Nov. 2006. [Online]. Available: <http://link.springer.com/10.1007/s10844-006-9953-7>
- [24] Q. Cai, L. Liao, F. Zou, S. Song, J. Liu, and M. Zhang, "Trajectory Similarity Measuring with Grid-Based DTW," in *Advances in Smart Vehicular Technology, Transportation, Communication and Applications*, ser. Smart Innovation, Systems and Technologies, Cham, 2019, pp. 63–72.
- [25] L. Ries, P. Rigoll, T. Braun, T. Schulik, J. Daube, and E. Sax, "Trajectory-Based Clustering of Real-World Urban Driving Sequences with Multiple Traffic Objects," in *2021 IEEE International Intelligent Transportation Systems Conference (ITSC)*, Sep. 2021, pp. 1251–1258.
- [26] L. Chen, M. T. Özsu, and V. Oria, "Robust and fast similarity search for moving object trajectories," in *Proceedings of the 2005 ACM SIGMOD international conference on Management of data-SIGMOD '05*. Baltimore, Maryland: ACM Press, 2005, p. 491.
- [27] S. Ranu, Deepak P, A. D. Telang, P. Deshpande, and S. Raghavan, "Indexing and matching trajectories under inconsistent sampling rates," in *2015 IEEE 31st International Conference on Data Engineering*. Seoul, South Korea: IEEE, Apr. 2015, pp. 999–1010.
- [28] S. Wang, Z. Bao, J. S. Culpepper, T. Sellis, and X. Qin, "Fast large-scale trajectory clustering," *Proceedings of the VLDB Endowment*, vol. 13, no. 1, pp. 29–42, Sep. 2019.
- [29] K. Zheng, Y. Zheng, N. J. Yuan, and S. Shang, "On discovery of gathering patterns from trajectories," in *2013 IEEE 29th International Conference on Data Engineering (ICDE)*, Apr. 2013, pp. 242–253, iSSN: 1063-6382.
- [30] T. Qian, S. Sun, X. Shan, X. Wei, C. Tai, and C. Liu, "Distributed-Swarm: A Real-Time Pattern Detection Model Based on Density Clustering," *IEEE Access*, vol. 10, pp. 59 832–59 842, 2022, conference Name: IEEE Access.
- [31] S. Wang, J. Cao, and P. Yu, "Deep Learning for Spatio-Temporal Data Mining: A Survey," *IEEE Transactions on Knowledge and Data Engineering*, pp. 1–1, 2020, conference Name: IEEE Transactions on Knowledge and Data Engineering.

- [32] C. Zhang, P. Patras, and H. Haddadi, "Deep Learning in Mobile and Wireless Networking: A Survey," *IEEE Communications Surveys Tutorials*, vol. 21, no. 3, pp. 2224–2287, 2019, conference Name: IEEE Communications Surveys Tutorials.
- [33] H. Zhang, X. Zhang, Q. Jiang, B. Zheng, Z. Sun, W. Sun, and C. Wang, "Trajectory Similarity Learning with Auxiliary Supervision and Optimal Matching," in *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence*. Yokohama, Japan: International Joint Conferences on Artificial Intelligence Organization, Jul. 2020, pp. 3209–3215. [Online]. Available: <https://www.ijcai.org/proceedings/2020/444>
- [34] X. Wu, R. Zhang, and L. Li, "Clustering Trajectories via Sparse Auto-encoders," in *2021 IEEE 4th International Conference on Multimedia Information Processing and Retrieval (MIPR)*, Sep. 2021, pp. 260–266.
- [35] X. Liu, X. Tan, Y. Guo, Y. Chen, and Z. Zhang, "CSTRM: Contrastive Self-Supervised Trajectory Representation Model for trajectory similarity computation," *Computer Communications*, vol. 185, pp. 159–167, Mar. 2022. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S0140366422000019>
- [36] Z. Mao, Z. Li, D. Li, L. Bai, and R. Zhao, "Jointly Contrastive Representation Learning on Road Network and Trajectory," in *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*. Atlanta GA USA: ACM, Oct. 2022, pp. 1501–1510. [Online]. Available: <https://dl.acm.org/doi/10.1145/3511808.3557370>
- [37] Z. Fang, Y. Du, X. Zhu, D. Hu, L. Chen, Y. Gao, and C. S. Jensen, "Spatio-Temporal Trajectory Similarity Learning in Road Networks," in *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, ser. KDD '22. New York, NY, USA: Association for Computing Machinery, Aug. 2022, pp. 347–356. [Online]. Available: <https://doi.org/10.1145/3534678.3539375>
- [38] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention Is All You Need," *arXiv:1706.03762 [cs]*, Dec. 2017, arXiv: 1706.03762. [Online]. Available: <http://arxiv.org/abs/1706.03762>
- [39] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," May 2019, arXiv: 1810.04805.
- [40] L. v. d. Maaten and G. Hinton, "Visualizing Data using t-SNE," *Journal of Machine Learning Research*, vol. 9, no. 86, pp. 2579–2605, 2008. [Online]. Available: <http://jmlr.org/papers/v9/vandermaten08a.html>
- [41] Z. Wu, S. Wang, J. Gu, M. Khabsa, F. Sun, and H. Ma, "CLEAR: Contrastive Learning for Sentence Representation," Dec. 2020, arXiv:2012.15466 [cs]. [Online]. Available: <http://arxiv.org/abs/2012.15466>
- [42] D. Zhang, F. Nan, X. Wei, S.-W. Li, H. Zhu, K. McKeown, R. Nallapati, A. O. Arnold, and B. Xiang, "Supporting Clustering with Contrastive Learning," in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Online: Association for Computational Linguistics, Jun. 2021, pp. 5419–5430. [Online]. Available: <https://aclanthology.org/2021.naacl-main.427>
- [43] L. Deng, Y. Zhao, Z. Fu, H. Sun, S. Liu, and K. Zheng, "Efficient Trajectory Similarity Computation with Contrastive Learning," in *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*. Atlanta GA USA: ACM, Oct. 2022, pp. 365–374. [Online]. Available: <https://dl.acm.org/doi/10.1145/3511808.3557308>
- [44] S. Shang, L. Chen, Z. Wei, C. S. Jensen, K. Zheng, and P. Kalnis, "Trajectory similarity join in spatial networks," *Proceedings of the VLDB Endowment*, vol. 10, no. 11, pp. 1178–1189, Aug. 2017. [Online]. Available: <https://doi.org/10.14778/3137628.3137630>
- [45] F. Liu and Y. Deng, "Determine the Number of Unknown Targets in Open World Based on Elbow Method," *IEEE Transactions on Fuzzy Systems*, vol. 29, no. 5, pp. 986–995, May 2021, conference Name: IEEE Transactions on Fuzzy Systems.



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